

Sprint Journey

ReCell – AI-Powered EV Battery Intelligence Platform

(Idea Planning Phase)

Day 1 – Understanding the Problem Statement and Initial Brainstorming

Our sprint journey began with an introductory meeting where the team gathered to carefully read and analyze the problem statement. The central theme revolved around **developing innovative solutions that accelerate the clean energy transition through technology-driven approaches**. Instead of immediately jumping into a specific idea, we first focused on understanding the broader **energy ecosystem and its interconnected sectors**.

During this discussion, we mapped out the major segments within the energy industry, including renewable energy generation, electric mobility, energy storage systems, grid infrastructure, and smart energy management technologies. Each team member contributed perspectives on how these sectors are evolving and what technological challenges they currently face.

We also discussed the global push toward electrification and sustainability, particularly how emerging technologies such as artificial intelligence, IoT sensors, and predictive analytics are increasingly being used to optimize energy systems. The goal of this initial session was to identify areas where **software-based solutions could create scalable impact without requiring heavy infrastructure investments**.

By the end of the day, we concluded that instead of focusing on a single solution immediately, we needed to **explore different domains within the energy sector in depth** before deciding where to build our startup concept.

Day 2 – Exploring Different Energy Sector Opportunities

On the second day, we divided our research efforts and began studying different sectors of the energy industry in detail. Each member took responsibility for researching one area, including renewable energy optimization, grid integration technologies, EV infrastructure, and energy storage solutions.

We reviewed articles, industry reports, and startup case studies to understand where innovation is happening and where gaps still exist. One of our key objectives during this stage was to identify **problems that were both technologically interesting and practically solvable by a startup team.**

Our discussions revealed that many challenges in the energy sector are still dominated by hardware and infrastructure, which often require large investments and regulatory approvals. Therefore, we started leaning toward sectors where **data-driven software solutions could create value without the need for large-scale physical infrastructure.**

By the end of the day, we had compiled a list of several promising areas within the energy sector that we could potentially explore further.

Day 3 – Deep Research on Grid Integration, Renewable Energy, and Electric Mobility

After reviewing our initial research findings, we shortlisted three promising domains that appeared both impactful and relevant to the problem statement:

1. Smart grid and grid integration technologies
2. Renewable energy optimization systems
3. Electric vehicles and battery management

We began analyzing each of these domains in depth. Grid integration initially seemed like a strong candidate because the increasing adoption of renewable energy requires intelligent management of electricity flow and grid stability. However, while studying this sector, we realized that grid-related solutions often involve **complex regulatory frameworks, long implementation cycles, and dependency on government infrastructure**.

This raised concerns about scalability and speed of implementation for a startup like ours.

In contrast, the electric vehicle ecosystem is expanding rapidly across the world. Governments are encouraging EV adoption, automotive companies are investing heavily in electrification, and new charging infrastructure is being built globally. This rapid growth suggested that **EV-related technologies may offer greater opportunities for innovation and faster scalability**.

By the end of the third day, we began shifting our attention more strongly toward the EV sector.

Day 4 – Finalizing the EV Battery Management Domain

On the fourth day, we held a long discussion comparing the potential of grid integration and electric mobility. While grid integration technologies remain important for the energy transition, we concluded that they require heavy infrastructure investment and collaboration with utilities or governments. This makes them difficult to implement quickly within the scope of a startup sprint.

Electric vehicles, on the other hand, present several clear technological challenges that can be addressed through software solutions. One of the most critical components of an EV is its **battery system**, which directly determines the vehicle's range, performance, safety, and lifespan.

We studied how Battery Management Systems (BMS) work. These systems monitor parameters such as voltage, current, and temperature to ensure that the battery operates safely. However, most current BMS solutions are

rule-based and reactive, meaning they respond only after problems occur rather than predicting them in advance.

This insight revealed a clear opportunity: **using AI-driven analytics to make battery systems smarter and more predictive.**

This led to the formation of our startup idea:

ReCell – an AI-powered battery intelligence platform designed to monitor battery health, predict failures, and optimize charging behavior.

Day 5 – Market Study and User Research Planning

After finalizing the EV Battery Management domain on the previous day, we realized that before building any solution we needed to validate whether the problems we identified were actually experienced by real EV users and industry professionals. Instead of relying only on secondary research and articles, we decided to directly gather insights from people who interact with EV technology in their daily lives.

We began the day by conducting a **structured market study** of the EV ecosystem. This involved reviewing reports on EV adoption, battery degradation trends, charging infrastructure challenges, and the growing importance of battery lifecycle management. During this process, we observed that while the EV market is expanding rapidly, battery performance issues such as range anxiety, charging inefficiencies, and battery degradation remain major concerns for users.

To validate these observations, we decided to conduct primary research. We designed a **Google Form survey** that included questions related to EV usage patterns, charging behavior, range reliability, battery health concerns, and the types of battery insights users would find valuable. The survey also asked respondents about the challenges they face in monitoring battery performance and what features they would expect in a battery intelligence platform.

Once the questionnaire was finalized, we began circulating it among EV users, technology enthusiasts, and professionals in the energy sector through various channels.

In parallel, we also started reaching out to professionals through **LinkedIn**, particularly individuals working in EV startups, battery technology companies, charging infrastructure companies, and energy consulting firms. Our objective was to understand real-world battery challenges from people who are directly involved in the EV ecosystem.

Day 6 – Early Insights from Surveys and Beginning Market Research

On the sixth day, our team began closely monitoring the responses coming in from the Google Form survey that we had circulated among EV users and energy sector enthusiasts. The responses started giving us early insights into the challenges faced by EV users, particularly regarding uncertainty about battery health, inaccurate range predictions, and concerns about long-term battery degradation.

While the survey responses were being collected, we also continued our outreach efforts on LinkedIn. Several professionals and EV enthusiasts responded to our messages and expressed interest in learning more about the idea. Some of them shared brief comments about their own experiences with EV battery performance and suggested areas where better battery analytics could be helpful.

Alongside collecting user feedback, we began conducting **market research to understand the EV ecosystem more clearly**. We studied reports about EV adoption trends, the growth of electric mobility markets, and the role of battery technology in determining vehicle performance. This research helped us understand that the EV ecosystem is expanding rapidly, but there are still significant gaps in battery lifecycle management and predictive analytics.

At the same time, we began drafting the **initial structure of our pitch deck**. The goal was to gradually build the presentation alongside our research so that the insights we gathered could directly strengthen our problem statement and solution narrative.

Day 7 – Competitor Analysis and Expanding Industry Outreach

On Day 7, we focused on understanding the **competitive landscape in the EV battery analytics space**. We studied several companies that operate in this domain, including battery analytics platforms, EV fleet management systems, and charging infrastructure software providers.

Through this competitor analysis, we discovered that many companies focus on **only one aspect of battery intelligence**, such as battery health monitoring or charging optimization. However, very few platforms integrate multiple capabilities such as predictive battery health analytics, charging optimization, fleet monitoring, and lifecycle management into a single system.

This observation helped us refine the positioning of ReCell as a **comprehensive battery intelligence platform** rather than just another monitoring tool.

Meanwhile, our LinkedIn outreach continued generating responses. Several professionals expressed enthusiasm about the idea and were willing to discuss the concept in more detail through short online meetings. These responses encouraged us to continue engaging with people in the EV ecosystem.

At the same time, we continued refining the **problem and market sections of our pitch deck**, incorporating insights from both survey responses and competitor analysis.

Day 8 – User Insights and Feature Refinement

By Day 8, we had received a considerable number of responses through the survey as well as feedback through LinkedIn messages. Many EV users highlighted the lack of transparency in current battery information systems. They explained that while dashboards display battery percentage, they do not provide meaningful insights about battery health or long-term performance.

These insights confirmed that battery intelligence platforms could provide real value by making battery data more accessible and actionable.

Based on these findings, we began refining the **feature set of ReCell**. Instead of building a platform with too many functionalities, we decided to focus on features that directly addressed the most common user concerns.

At the same time, we also began working on the **product section of the pitch deck**, explaining how our proposed platform would solve these problems.

Day 9 – Expert Discussions and Technical Validation

On the ninth day, we conducted discussions with individuals who had shown interest in speaking with us through LinkedIn. These conversations helped us understand the operational and technical challenges involved in battery management.

Several professionals mentioned that traditional battery management systems primarily monitor parameters like voltage and temperature but often lack **predictive analytics that can anticipate battery degradation or failure**.

We also held a meeting with professors from the **Energy Department**, who provided technical insights into battery systems and their limitations. They explained how factors such as charging rate, temperature fluctuations, and usage patterns influence battery degradation.

These discussions helped us validate the feasibility of our solution and refine our understanding of how predictive analytics could improve battery lifecycle management.

At the same time, we continued working on the **technical explanation and architecture sections of our pitch deck**, incorporating the insights we had gathered.

Day 10 – Market Sizing and Business Model Exploration

After gaining a deeper understanding of the problem and the existing solutions in the market, we shifted our focus toward analyzing the **market opportunity for battery intelligence platforms**.

We studied reports related to EV adoption, battery analytics markets, and EV fleet management software. Through this research, we estimated the potential market size for platforms that provide battery monitoring, predictive analytics, and charging optimization services.

This analysis helped us define the **TAM, SAM, and SOM** for our startup concept.

At the same time, we began discussing possible **business models** for ReCell. We explored options such as SaaS subscriptions for fleet operators, API integrations for EV manufacturers, and analytics dashboards for charging infrastructure providers.

These discussions helped us shape the **business model and market opportunity sections of our pitch deck**.

Day 11 – Product Design and Prototype Wireframes

Once the problem, market opportunity, and feature set were clearly defined, we moved on to visualizing how the platform would function in practice.

We started designing **prototype wireframes** that represented the interface of the ReCell platform. These wireframes illustrated how users would interact with the system and how battery insights would be presented.

Some of the prototype screens included:

- Battery health dashboard
- Predictive failure alerts
- Charging optimization recommendations

- Fleet analytics overview

Creating these wireframes helped us translate our conceptual ideas into a more tangible product design. It also helped us identify areas where the user experience could be simplified.

At the same time, we incorporated these visuals into the **product section of our pitch deck**.

Day 12 – Final Consolidation and Pitch Preparation

On the final day of the sprint journey, we consolidated everything we had learned and developed during the previous days.

We reviewed our survey responses, industry discussions, market research findings, and prototype designs to ensure that all aspects of our solution were well aligned.

The final concept of ReCell was defined as:

An AI-powered battery intelligence platform that helps EV users, fleet operators, and manufacturers monitor battery health, predict failures, optimize charging behavior, and extend battery lifespan.

We finalized the pitch deck by integrating insights from our research, user feedback, competitor analysis, and prototype design.

By the end of the sprint journey, we had successfully transformed an initial exploration of the energy sector into a **well-structured startup concept supported by research, industry insights, and a clear product vision**.



